Process Characterization Report

A-Mab Drug Substance — Anion Exchange Chromatography (Step 8) — PCR-008

Process Development / Manufacturing Science & Technology (MSAT)

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Abbreviations

AEX, anion-exchange chromatography · ANOVA, analysis of variance · CCD, central-composite design · CEX, cation-exchange chromatography · CPP, critical process parameter · CQA, critical quality attribute · CV, column volume · DoE, design of experiments · DS, drug substance · ELISA, enzyme-linked immunosorbent assay · FT, flow-through · GPP, general process parameter · HCP, host-cell protein · icIEF, imaged capillary isoelectric focusing · ICH, International Council for Harmonisation · IPC, in-process control · KPP, key process parameter · LRF/LRV, log-reduction factor / value · MSAT, Manufacturing Science and Technology · MVM, minute virus of mice (parvovirus) · NOR, normal operating range · PAR, proven acceptable range · PPQ, process performance qualification · qPCR, quantitative polymerase chain reaction · QbD, Quality by Design · RSM, response-surface methodology · SDM, scale-down model · TCID50, 50% tissue-culture infectious dose · UV, ultraviolet absorbance (280 nm) · WC-CPP, well-controlled CPP · XMuLV, xenotropic murine leukaemia virus (enveloped model virus).

Executive summary

This report documents the Stage 1 (Process Design) characterization of the A-Mab anion-exchange (AEX) polishing chromatography step (Step 8), the **flow-through** polishing operation that follows cation-exchange polishing and is the **final chromatographic step** of the purification train. Operated in flow-through (product-transmission) mode, anion exchange is the step at which the process **establishes the minute-virus-of-mice (MVM, parvovirus) viral-clearance claim** — the CQA it sets — and it provides orthogonal enveloped-virus (XMuLV) clearance and a major further reduction of host-cell protein (HCP), with modular clearance of residual DNA and leached Protein A. The study was executed under the Process Characterization Plan **PCP-008** on a qualified scale-down chromatography column with a qualified small-scale viral-spiking model, following the lifecycle approach to process validation (U.S. Food and Drug Administration 2011; Parenteral Drug Association 2013; International Society for Pharmaceutical Engineering 2023), the Quality-by-Design principles of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023b, 2012), and the viral-safety framework of ICH Q5A(R2) (International Council for Harmonisation 2023a), with the process model, quality-attribute framework and risk methodology of the A-Mab case study (CMC Biotech Working Group 2009).

Five process parameters were evaluated. A 19-run two-level full factorial screening design identified the significant factors and two-factor interactions for the flow-through-pool HCP, the XMuLV and MVM log-reduction and the step yield; a 28-run face-centred central-composite response-surface design then quantified main effects, interactions and curvature and defined the multivariate operating region in the well-controlled CPPs. The flow-through-pool HCP, XMuLV-clearance and MVM-clearance response-surface models are adequate (R^2 = 0.90–0.93, adjusted R^2 0.79, no significant lack of fit); the step yield is robust across the ranges studied.

Two deviations from PCP-008 were recorded and resolved (§12). First, the load material used in the initial execution of the designs was found to carry an elevated acidic charge-variant (deamidation) burden that, through an anomalous protein-load × Equil/Wash-1-conductivity interaction, drove the flow-through-pool purity to unacceptable levels; the affected designs were invalidated and re-executed in full on a requalified, representative load — the re-executed designs are the ones reported here, and in them that interaction is statistically absent, confirming it was an artifact of the degraded load. Second, both executions used a trailing-edge UV pool-collection set-point that was subsequently found to be slightly permissive; because insufficient requalified load remained for a third design, the corrected pool-stop was determined by modelling the recorded chromatographic data and confirmed with a justified set of verification runs. Neither deviation altered the parameter classifications or the operating region established by the DoE.

Key outcomes

- All four multivariate parameters significantly affect a product-quality CQA and are classified **well-controlled CPP (WC-CPP)**: load pH and the equilibration/wash-1 conductivity drive the flow-through-pool HCP through a significant load-pH × conductivity interaction, and load pH and the load conductivity drive the XMuLV and MVM log-reduction. The protein load has no significant effect on clearance in representative load material and is controlled as a WC-CPP with a wide range; the operating flow rate is a WC-CPP controlled univariately to preserve the residence time credited for viral clearance. There are no key process parameters (KPP) at this step.

- Anion exchange sets the cumulative **MVM (parvovirus) clearance** claim, contributing approximately **4.7 log** at the nominal condition — the first and principal MVM-clearance step — so that the drug substance meets its 8.6 log cumulative MVM requirement (achieved cumulative **10.0 log** , Cpk = 1.51).
- The step contributes approximately **6.0 log** of orthogonal enveloped-virus (XMuLV) clearance toward the 16.7 log cumulative requirement (achieved cumulative **18.9 log** , Cpk = 1.53), and reduces the pool HCP by approximately **6.2-fold** to about **19 ng/mg**, below the 100 ng/mg drug-substance limit.
- The nominal step recovers **97.3%** of the loaded product (flow-through, high recovery) and the yield is robust across the ranges studied.
- A multivariate operating region in the four well-controlled CPPs was established over which the flow-through-pool HCP and the credited MVM and XMuLV log-reduction are controlled so that the integrated control strategy delivers in-specification drug substance. The MVM clearance capability (**Cpk = 1.51**) is the tightest process capability of the A-Mab drug substance, and anion exchange is its principal contributor.

The anion-exchange step is demonstrated to be well understood and robust. The operating region, parameter classifications, proven acceptable ranges and the modular viral-clearance credit support the Stage 2 operating ranges and the drug-substance control strategy, and this report rolls up into the Process Characterization Master Report (**PCMR-001**).

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody purified by a platform Protein A → low-pH viral-inactivation → cation-exchange → anion-exchange → virus-filtration → UF/DF train (CMC Biotech Working Group 2009). Anion-exchange chromatography is the second (final) polishing step and is operated in **flow-through** mode: the cation-exchange eluate is conditioned to the target load pH and load conductivity and loaded onto an equilibrated strong anion-exchange resin; at the operating pH the product antibody carries a slight net positive charge and transmits through the column, while the more acidic process-related impurities — host-cell protein, residual DNA and endotoxin — and the model viruses bind and are retained. The product is recovered in the flow-through together with a post-load wash, the pool being collected from the start of load to a defined trailing-edge UV (280 nm) pool-stop criterion. The step forms no new size- or glycan-related CQA; its role in the control strategy is to **credit the MVM (parvovirus) log-reduction** required for the drug-substance viral-safety claim, to provide orthogonal enveloped-virus (XMuLV) clearance, and to complete the reduction of HCP with modular clearance of residual DNA and leached Protein A. This characterization quantifies how the anion-exchange parameters govern the flow-through-pool HCP, the XMuLV and MVM log-reduction and the step yield.

1.2 Objectives

The objectives of the study were to (i) establish the quantitative relationships between the anion-exchange process parameters and the flow-through-pool HCP, the XMuLV and MVM log-reduction and the step yield; (ii) define the multivariate operating region over which the flow-through-pool HCP and the credited viral clearance are controlled; (iii) establish proven acceptable ranges (PARs) and normal operating ranges (NORs) for each parameter; (iv) classify each parameter for its risk of impact to CQAs and to process performance; and (v) provide the process understanding and ranges that justify the Stage 2 operating conditions, the modular viral-clearance claim and the control strategy.

1.3 Regulatory and scientific basis

The work follows the lifecycle approach to process validation of the FDA 2011 guidance (U.S. Food and Drug Administration 2011), PDA Technical Report 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023), applies the enhanced (QbD) development approach of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023b, 2012), and follows the viral-safety evaluation framework of

ICH Q5A(R2) (International Council for Harmonisation 2023a) for the modular clearance claim. Criticality is treated as a continuum (ICH Q9). The report structure corresponds to the Stage 1 Process Design Report of PDA TR 60 §3.11 and provides development-report content suitable to support a Biologics License Application.

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Flow-through anion-exchange polishing is a platform operation whose behaviour is well established from related humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and from A-Mab clinical and engineering experience (CMC Biotech Working Group 2009). Prior knowledge established that, in flow-through mode, impurity and virus binding is governed by the electrostatic environment: a higher load pH increases the net negative charge of the impurities and virus, strengthening their retention on the resin and improving clearance (lower flow-through-pool HCP, higher log-reduction), while a higher equilibration/wash-1 conductivity screens those electrostatic interactions and weakens retention, reducing HCP clearance — the two acting together through a load-pH × conductivity interaction; that viral clearance additionally requires a sufficiently low load conductivity for the virus to bind, so that clearance falls as the load conductivity rises; that, because the operation is flow-through with high dynamic capacity for the product, the protein load can be run high with little effect on clearance or recovery over the platform range provided the load is of representative charge-variant quality; and that residence time, set by the operating flow rate, governs the contact time available for binding and is held within a range that preserves the minimum contact time credited for the viral-clearance claim. This knowledge framed the attributes in scope and the prioritization of parameters for characterization.

2.2 Quality attributes in scope

Anion exchange **sets one CQA of its own** — the cumulative MVM (parvovirus) log-reduction, to which it is the first and principal contributor — and is a major clearance step for the enveloped-virus (XMuLV) log-reduction and for the process-related impurity CQAs formed upstream and specified at the drug substance: host-cell protein, residual DNA and leached Protein A (Table 2.1). Criticality is scored on the A-Mab continuum using Tool #1 (Risk Score = Impact × Uncertainty); attributes rated High (H) or Very High (VH) are treated as *Critical* (CMC Biotech Working Group 2009).

Table 2.1: CQAs in scope for the anion-exchange step — the MVM clearance it sets and the XMuLV clearance and impurity CQAs it controls/clears — with acceptance criterion and criticality (Tool #1 = Impact × Uncertainty).

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20

CQA	Category	Acceptance	Criticality	Tool #1
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16

The MVM and XMuLV log-reductions are the **cumulative** drug-substance viral-clearance claims (8.6 and 16.7 log respectively), to which the anion-exchange step contributes a modular increment quantified here; the flow-through-pool HCP is an in-process attribute whose control at this step determines the margin to the 100 ng/mg HCP drug-substance limit. The log-reduction credited to the step, and the flow-through-pool HCP, are therefore the primary responses of the study; residual DNA and leached Protein A are measured to confirm the modular clearance credited across Protein A, cation exchange and anion exchange. The load-material acidic charge-variant (deamidation) profile — a low-criticality attribute with a 18–40% design-space range — is a controlled input attribute of the feed; its relevance to this step is discussed with the deviations (§12).

2.3 Risk-based prioritization and parameter selection

Consistent with the A-Mab lifecycle of iterative risk assessments, the characterization scope was set by the Pre-Characterization Process Risk Assessment (**RA-001**), which used prior knowledge and a risk-ranking-and-filtering screen to prioritize parameters and to assign the study type. Load pH, equilibration/wash-1 conductivity, load conductivity and protein load, which carry a credible main-effect and interaction risk to the flow-through-pool HCP or to the viral log-reduction, were assigned to the multivariate DoE; the operating flow rate, which acts on the viral-clearance claim through residence time rather than through the buffer/load chemistry, was assigned to univariate assessment and controlled to protect the minimum contact time (CMC Biotech Working Group 2009; Parenteral Drug Association 2013). The parameters and ranges carried into this study are given in Table 4.1 (§5.1).

3 Materials and methods

3.1 Scale-down model and its qualification

All characterization runs were performed on a qualified scale-down anion-exchange column operated at the commercial-process equivalents for resin, bed height, residence time, buffer system, load ratio and pool-collection strategy (**SOP-2011**, **SOP-2008**); the XMuLV and MVM log-reduction were determined under a dedicated qualified small-scale viral-spiking study using qualified virus stocks (International Council for Harmonisation 2023a). The SDM was qualified against at-scale reference data (**SOP-1001**) by comparing the operational parameters and, critically, the input and output attributes — step yield, flow-through-pool HCP, residual DNA and leached Protein A, and the XMuLV and MVM log-reduction — confirming that the model is representative and suitable to support operating-range, PAR and viral-clearance claims (Parenteral Drug Association 2013). The between-run reproducibility of the centre-point condition (§6.1) provides a direct estimate of model precision and is used as the pure-error term in the lack-of-fit assessment of every response-surface model.

The screening and response-surface designs were **executed twice**. As documented in §12 (Deviation DEV-01), the first execution used a load lot subsequently found to be non-representative — it carried an elevated acidic charge-variant (deamidation) burden — and was invalidated for operating-region definition; the designs were re-executed in full on a requalified, representative load. **The re-executed designs are the ones analyzed and reported here.**

3.2 Operation

The cation-exchange eluate is conditioned to the target load pH and load conductivity and loaded onto the equilibrated anion-exchange column; the product transmits in the flow-through while impurities and virus bind; and a post-load wash is applied at the equilibration/wash-1 conductivity. The flow-through/wash pool is collected from the start of load to a defined trailing-edge UV (280 nm) pool-stop criterion. Buffers, resin lot and column packing are controlled to platform specifications (**SOP-2008**) and are treated as identity/quality-controlled inputs rather than characterized variables (CMC Biotech Working Group 2009). The charge-variant quality of the load material is verified prior to loading.

3.3 Analytical methods

The step responses were measured by validated methods (Table 3.1). Host-cell protein was determined by ELISA (**AMV-3012**); the enveloped-virus (XMuLV) and parvovirus (MVM) log-reduction by infectivity titre from the small-scale spiking study (**AMV-3017**, **AMV-3018**); residual DNA by qPCR (**AMV-3014**); leached Protein A by ELISA (**AMV-3016**); and the load-material and pool acidic/deamidated charge-variant profile by imaged capillary isoelectric focusing (**AMV-3013**). Step yield was determined from the load and flow-through-pool product mass by A . Method performance characteristics are summarized in Appendix D.

Table 3.1: Validated analytical methods used in the characterization.

Reference	Title	Type
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

3.4 Sampling plan

Each DoE run was sampled at the flow-through pool for HCP, residual DNA and leached Protein A, and the load and pool product mass were recorded for the step-yield determination; the paired viral-spiking runs provided the XMuLV and MVM log-reduction, and the load-material acidic-variant profile was recorded for each execution. In-process readings (pH, conductivity, UV) confirmed the chromatographic profile and the pool collection window. Centre-point runs were replicated within each design (three in screening, four in the response-surface design) to quantify reproducibility and pure error.

3.5 Statistical methods

Designs were constructed and analyzed in coded factors, with each parameter's characterization range mapped so that the set-point is 0 and the range edges are ± 1 (**SOP-1002**). Screening data were analyzed by ordinary-least-squares regression of each response on the main effects and all two-factor interactions; factor *effects* are reported as twice the coded coefficient. Response-surface data were analyzed with the full quadratic model (main effects, two-factor interactions and pure-quadratic terms). Statistical significance was assessed at $= 0.05$. Model adequacy was judged by the coefficient of determination (R^2), the adjusted R^2, the predicted R^2 (computed from the PRESS statistic by leave-one-out), the overall model F-test, and an analysis of variance partitioning the residual into lack of fit and pure error, the latter estimated from the replicated centre points; a non-significant lack-of-fit F-test ($p > 0.05$) indicates an adequate model. A response governed by no active factor over the ranges studied is treated as robust rather than as a predictive surface. The operating region was defined as the multivariate region of the well-controlled CPPs over which the flow-through-pool HCP and the credited MVM and XMuLV log-reduction are controlled. Commercial-scale process capability for the drug-substance CQAs governed by the step was estimated by Monte-Carlo simulation of 2,000 batches with each parameter varied within its NOR, reporting the one-sided process-capability index (Cpk) — against the upper limit for the impurity CQAs and against the lower requirement for the cumulative viral-clearance CQAs.

4 Study design

4.1 Factors, ranges and the knowledge space

Five parameters were studied (Table 4.1). The characterization range of each factor ("PAR" column) defines the edges of the explored knowledge space and the basis of the proven acceptable range; the NOR is the tighter range at which the parameter is controlled routinely. The DoE factor coding for the four multivariate factors is given in Table 4.2.

Table 4.1: Anion-exchange process parameters, set-points, ranges and post-characterization classification.

Parameter	Unit	Set-point	NOR	PAR	Class	Study
Load pH	pH	7.5	7.35–7.65	7.2–7.8	WC-CPP	multivariate
Equil/Wash-1 conductivity	mS/cm	2.6	2.1–3.1	1.6–3.6	WC-CPP	multivariate
Load conductivity	mS/cm	5.5	4–7	3–8	WC-CPP	multivariate
Protein load	g/L resin	200	120–280	50–300	WC-CPP	multivariate
Operating flow rate	cm/hr	300	200–400	150–450	WC-CPP	univariate

Table 4.2: Design-of-experiments factor legend (coded A–D).

Code	Factor	Unit	Studied in
A	Load pH	pH	screening + RSM
B	Equil/Wash-1 conductivity	mS/cm	screening + RSM
C	Load conductivity	mS/cm	screening + RSM
D	Protein load	g/L resin	screening + RSM

4.2 Screening design

Factor screening used a two-level full factorial in the four multivariate factors (A–D) with three centre-point replicates, 19 runs in total. The full factorial estimates all four main effects and all six two-factor interactions without confounding. The four responses recorded for each run were the flow-through-pool HCP, the XMuLV log-reduction, the MVM log-reduction and the step yield. The complete coded design matrix and run responses are given in Appendix A.

4.3 Response-surface design

The four multivariate factors were carried into a face-centred central-composite design (cube points, axial points at the face centres, and four centre-point replicates), 28 runs in total, enabling estimation of curvature in addition to main effects and interactions. The complete coded design matrix and responses are given in Appendix B.

4.4 Univariate assessment

The operating flow rate was evaluated one factor at a time across its characterization range for its effect on the flow-through-pool HCP, the XMuLV and MVM log-reduction and the step yield; it was not carried into the multivariate design because it acts on viral clearance through residence time rather than through the buffer/load chemistry and is not expected to interact with the chromatographic factors.

5 Results

5.1 Centre-point performance and reproducibility

At the nominal (centre-point) condition the step recovers 96.6% of the loaded product and delivers a flow-through-pool HCP of approximately 14 ng/mg, an XMuLV log-reduction of approximately 6.6 log and an MVM log-reduction of approximately 4.9 log . The between-replicate reproducibility of the centre point (Table 5.1) is the pure-error estimate used in the lack-of-fit assessment; the coefficients of variation reflect the combined analytical and process variability of the scale-down and viral-spiking models, with the pool-HCP measurement the least precise of the responses.

Table 5.1: Centre-point reproducibility (response-surface design; n = 4 replicates).

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	4	13.7	2.48	18.2
XMuLV LRF (log)	4	6.63	0.445	6.71
MVM LRF (log)	4	4.9	0.336	6.86
Step yield	4	0.966	0.0123	1.27

5.2 Screening: factor effects

The screening models resolve the active factors and interactions for the flow-through-pool HCP and the XMuLV and MVM log-reduction, and confirm that no factor systematically affects the step yield over the ranges studied (Table 5.2). Consistent with a screening analysis, the magnitudes are refined by the response-surface models (§6.3).

Table 5.2: Screening-design model summary by response (main effects + two-factor interactions).

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Pool HCP (ng/mg)	19	0.966	0.923	0.841	22.6	8.55e-05	2.45
XMuLV LRF (log)	19	0.842	0.644	-0.106	4.25	0.0257	0.393
MVM LRF (log)	19	0.97	0.932	0.869	25.8	5.18e-05	0.178
Step yield	19	0.498	-0.13	-3.64	0.792	0.642	0.0126

The largest effects are tabulated for the flow-through-pool HCP (Table 5.3) and the MVM log-reduction (Table 5.4). Full effect tables for all four responses are given in Appendix C.

Table 5.3: Largest standardized effects on flow-through-pool HCP (screening; effect = 2 × coded coefficient; A = load pH, B = Equil/Wash-1 conductivity, C = load conductivity, D = protein load).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	12.1	6.04	0.613	9.85	9.48e-06	***
A	-11.9	-5.96	0.613	-9.72	1.05e-05	***
AB	-6.63	-3.32	0.613	-5.41	0.000642	***
C	-1.66	-0.829	0.613	-1.35	0.213	
AC	1.65	0.827	0.613	1.35	0.214	
BC	-1.21	-0.605	0.613	-0.986	0.353	
D	0.885	0.443	0.613	0.722	0.491	
AD	-0.498	-0.249	0.613	-0.406	0.696	

Table 5.4: Largest standardized effects on MVM log-reduction (screening).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.22	0.61	0.0446	13.7	7.83e-07	***
C	-0.715	-0.357	0.0446	-8.01	4.32e-05	***
B	0.182	0.091	0.0446	2.04	0.0757	
BD	0.107	0.0536	0.0446	1.2	0.263	
AD	-0.0627	-0.0314	0.0446	-0.704	0.502	
D	-0.0451	-0.0226	0.0446	-0.506	0.626	
BC	-0.0442	-0.0221	0.0446	-0.496	0.634	
CD	0.0334	0.0167	0.0446	0.375	0.718	

The flow-through-pool HCP is governed by the load pH and the equilibration/wash-1 conductivity, which act in opposite directions — a higher load pH lowers the pool HCP (better clearance) and a higher wash-1 conductivity raises it (poorer clearance) — with a significant load-pH × conductivity interaction (effect -6.6 ng/mg, $p < 0.001$) that means the conductivity matters most at low load pH. Notably, the protein load and the protein-load × conductivity interaction have **no** significant effect on the pool HCP in this representative-load DoE (the protein-load × conductivity interaction effect is 0.22 ng/mg, $p = 0.86$); this is central to the resolution of Deviation DEV-01 (§12) and to the classification of the protein load. The XMuLV and MVM log-reductions are both governed by the load pH (clearance improves as pH rises) and the load conductivity (clearance falls as conductivity rises), with the load-pH main effect the largest for each; the wash-1 conductivity and the protein load have little effect on the viral log-reduction. The step yield shows no significant dependence on any factor, consistent with a high-recovery flow-through operation. The HCP and viral responses therefore respond to a partly shared set of factors (load pH is common), which is why a multivariate operating region — rather than independent one-dimensional ranges — is used to assure the impurity and viral load.

5.3 Response-surface models

The full quadratic response-surface models for the flow-through-pool HCP and the XMuLV and MVM log-reduction are adequate (Table 5.5): R^2 = 0.90–0.93, adjusted R^2 0.79. For all three modelled responses the lack-of-fit F-test was not significant ($p > 0.05$) relative to the centre-point pure error, confirming model adequacy. The pool-HCP and XMuLV models are additionally predictive (predicted R^2 0.62); the MVM model is adequate (R^2 = 0.90, adjusted R^2 = 0.79, no significant lack of fit) with a more modest predicted R^2 (0.39) that reflects the narrower log-reduction range of the parvovirus response relative to the spiking-assay reproducibility, and is corroborated by its strong, highly significant load-pH and load-conductivity main effects. The step-yield surface carries no active factor and is reported as robust across the ranges studied.

Table 5.5: Response-surface model summary by response (full quadratic model).

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Pool HCP (ng/mg)	28	0.934	0.864	0.66	13.2	1.83e-05	2.65
XMuLV LRF (log)	28	0.915	0.824	0.621	10	8.69e-05	0.314
MVM LRF (log)	28	0.898	0.787	0.394	8.14	0.000269	0.32
Step yield	28	0.488	-0.0632	-2.18	0.885	0.589	0.0145

The fitted coefficients for the flow-through-pool HCP and the MVM log-reduction are given in Table 5.6 and Table 5.7, and the analysis of variance for the flow-through-pool HCP — including the lack-of-fit/pure-error partition — in Table 5.8.

Table 5.6: Response-surface coefficients for flow-through-pool HCP (coded factors A = load pH, B = Equil/Wash-1 conductivity, C = load conductivity, D = protein load).

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	14.8	0.917	16.2	5.51e-10	***
A	-5.34	0.625	-8.54	1.09e-06	***
B	6.19	0.625	9.89	2.04e-07	***
C	0.0399	0.625	0.0637	0.95	
D	0.387	0.625	0.619	0.546	
AB	-1.45	0.663	-2.19	0.0475	*
AC	-0.925	0.663	-1.39	0.187	
AD	0.0192	0.663	0.029	0.977	
BC	0.15	0.663	0.226	0.825	
BD	-0.559	0.663	-0.843	0.415	
CD	-0.124	0.663	-0.187	0.855	
A^2	1.91	1.65	1.16	0.268	
B^2	2.22	1.65	1.34	0.202	
C^2	-0.0187	1.65	-0.0113	0.991	
D^2	-1.98	1.65	-1.2	0.253	

Table 5.7: Response-surface coefficients for MVM log-reduction.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	4.85	0.111	43.9	1.63e-15	***
A	0.539	0.0754	7.15	7.48e-06	***
B	0.0307	0.0754	0.407	0.691	
C	-0.564	0.0754	-7.48	4.65e-06	***
D	0.0261	0.0754	0.346	0.735	
AB	0.127	0.08	1.59	0.136	
AC	-0.074	0.08	-0.925	0.372	
AD	0.0353	0.08	0.441	0.666	
BC	-0.0339	0.08	-0.424	0.679	
BD	0.00805	0.08	0.101	0.921	
CD	-0.0156	0.08	-0.195	0.848	
A^2	-0.322	0.199	-1.61	0.13	
B^2	0.149	0.199	0.747	0.469	
C^2	0.0224	0.199	0.112	0.912	
D^2	0.135	0.199	0.679	0.509	

Table 5.8: Analysis of variance for the flow-through-pool HCP response-surface model, with lack-of-fit and pure-error partition.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	1.3e+03	14	93.1	13.2	1.83e-05
Residual	91.5	13	7.04	nan	nan
Lack of fit	73	10	7.3	1.19	0.499
Pure error	18.5	3	6.16	nan	nan

The flow-through-pool HCP and XMuLV response surfaces are shown in Figure 5.1 — pool HCP over load pH × equilibration/wash-1 conductivity (its dominant pair) and XMuLV log-reduction over load pH × load conductivity (the viral-clearance pair). The response surfaces for all four responses over load pH × equilibration/wash-1 conductivity are shown in Figure 5.2, and the model diagnostics for the pool-HCP, XMuLV and MVM models in Figure 5.3, Figure 5.4 and Figure 5.5. The residual-versus-predicted and normal-quantile plots show no systematic structure or departure from normality, and the actual-versus-predicted plots lie on the identity line, supporting the use of the models for the operating-region definition.

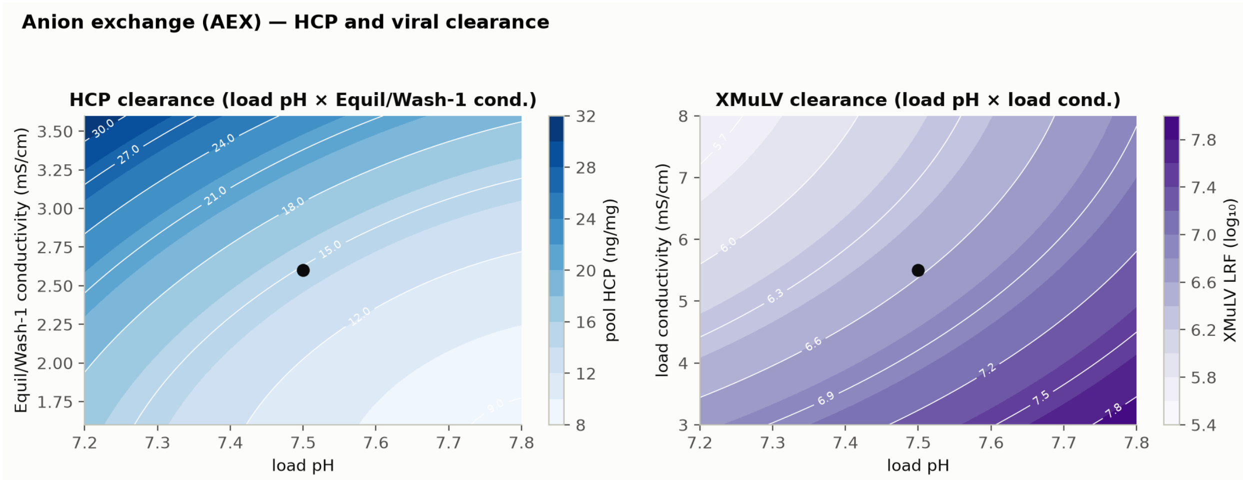

Figure 5.1: Anion-exchange response surfaces: flow-through-pool HCP falls as load pH rises and as equilibration/wash-1 conductivity falls, through a significant load-pH × conductivity interaction (left); XMuLV log-reduction falls as load pH falls and as load conductivity rises (right). The set-point is marked.

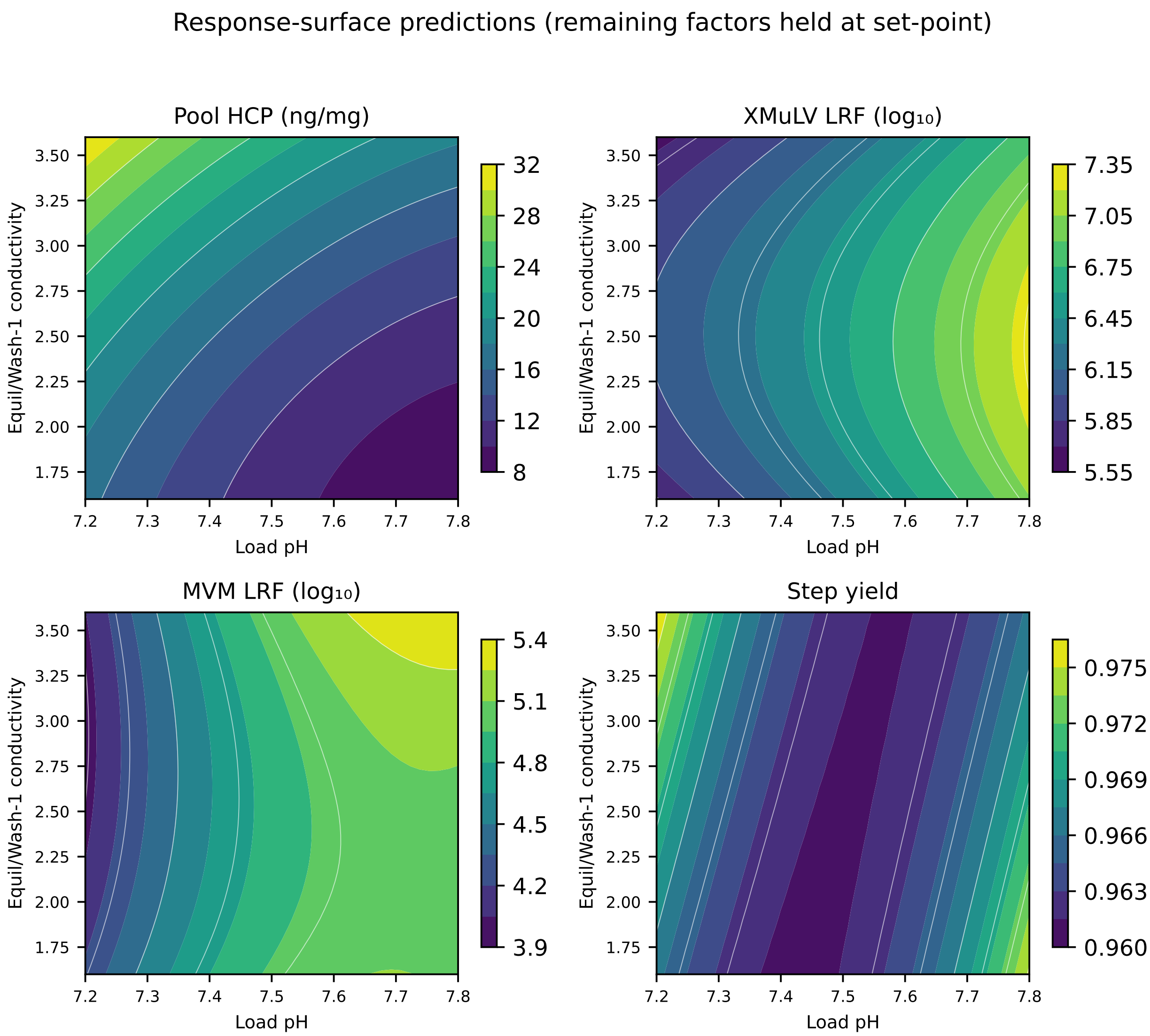

Figure 5.2: Response-surface predictions for the flow-through-pool HCP, XMuLV and MVM log-reduction and step yield over load pH × equilibration/wash-1 conductivity (remaining factors at set-point). Pool HCP responds strongly to both; the viral log-reductions respond to load pH; yield is flat.

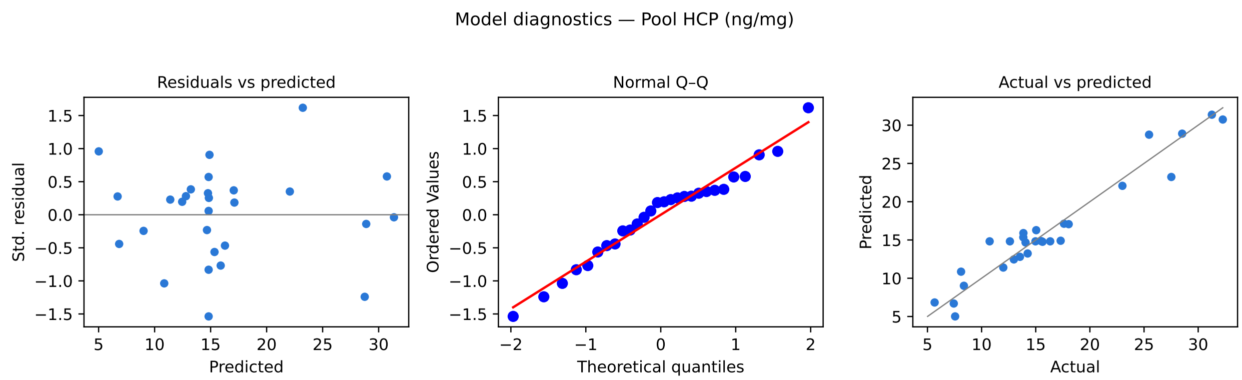

Figure 5.3: Model diagnostics for the flow-through-pool HCP response-surface model.

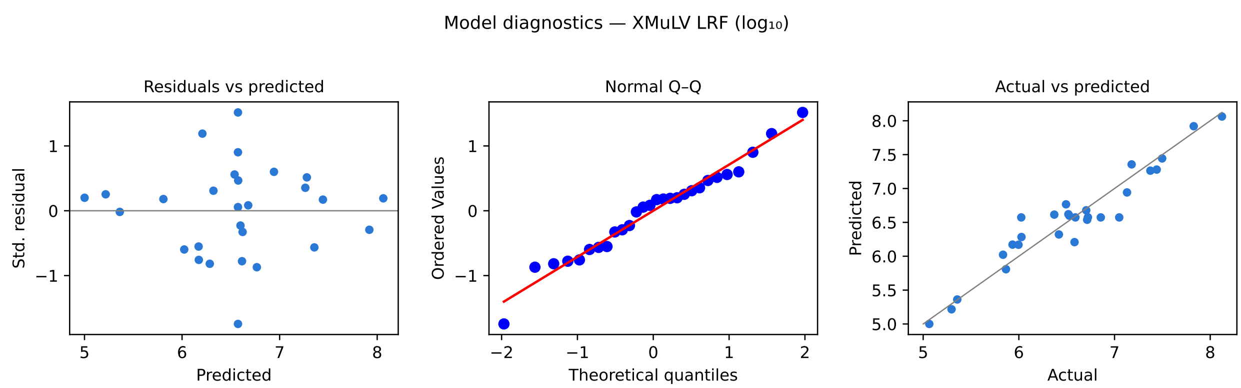

Figure 5.4: Model diagnostics for the XMuLV log-reduction response-surface model.

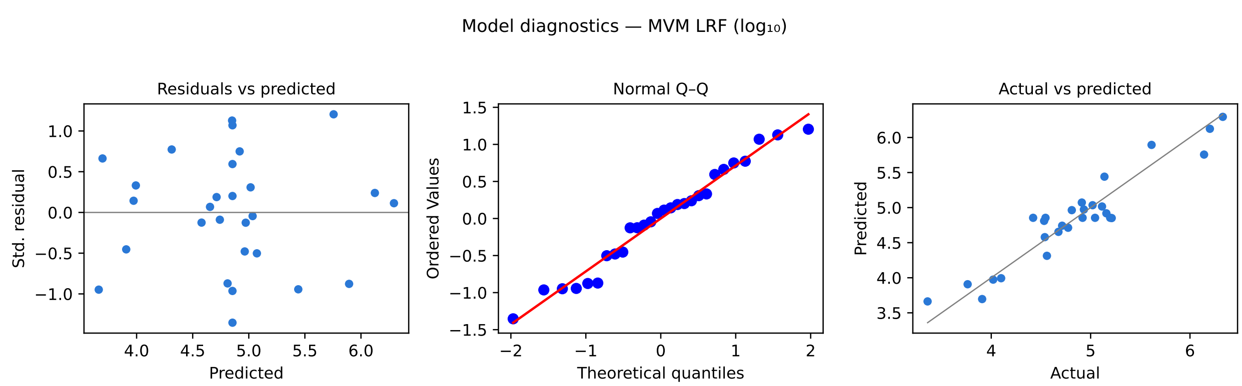

Figure 5.5: Model diagnostics for the MVM log-reduction response-surface model.

5.4 Mechanistic interpretation

The response-surface models reproduce the established behaviour of flow-through anion-exchange polishing. At the operating pH the antibody carries a slight net positive charge and transmits, while the more acidic host-cell protein, DNA and model viruses bind the positively charged resin; the strength of that binding — and therefore the clearance — is set by the electrostatic environment. A higher load pH increases the net negative charge of the impurities and virus, strengthening retention and improving clearance, so the flow-through-pool HCP falls and the XMuLV and MVM log-reductions rise; a higher equilibration/wash-1 conductivity screens the electrostatic interaction and weakens retention, raising the pool HCP, and the two act together through the significant load-pH $\times$ conductivity interaction, so that adequate load pH matters most when the wash conductivity is high. Viral clearance additionally requires a sufficiently low load conductivity for the virus to bind, so the XMuLV and MVM log-reductions fall as the load conductivity rises. Because the operation is flow-through with high dynamic capacity for the product, the protein load has no significant effect on clearance or recovery over the range studied when the load is of representative charge-variant quality — a finding that both widens the acceptable load range and, by contrast with the invalidated deamidated-load execution (§12), localizes the load-related risk to the charge-variant quality of the feed rather than to a load set-point. The step yield is high and flat, as expected for flow-through. This is the relationship depicted in Figure 5.1 and is the reason a multivariate operating region — rather than independent one-dimensional ranges — is used to assure the impurity and viral load presented downstream.

6 Design space

The operating region (design space) is the multivariate region of the four well-controlled CPPs over which the flow-through-pool HCP is held low enough, and the MVM and XMuLV log-reductions credited to the step are held high enough, that the drug substance meets its 100 ng/mg HCP limit and its 8.6 and 16.7 log cumulative viral-clearance requirements, while the step yield remains high. Its two principal planes are load pH × equilibration/wash-1 conductivity, over which the pool HCP is controlled (Figure 5.1, Figure 5.2), and load pH × load conductivity, over which the credited XMuLV and MVM log-reductions are controlled (Figure 5.1). The NOR of each parameter (Table 4.1) — load pH 7.35–7.65, equilibration/wash-1 conductivity 2.1–3.1 mS/cm, with the load conductivity and protein load at their NORs — lies well inside the characterized region, which in turn lies within the broader knowledge space bounded by the proven acceptable ranges. Because the protein load has no significant effect on clearance in representative material, the load axis of the region is wide (NOR 120–280 g/L resin); the load-related risk identified in the characterization is not a load set-point but the **charge-variant quality of the load material**, which is controlled as a feed-input attribute (§12, §11). Movement within the operating region is not a change, whereas movement outside it would require assessment under the quality system (CMC Biotech Working Group 2009; International Council for Harmonisation 2009). Because the step yield is robust and the HCP and viral responses are governed by a partly shared set of factors, the region is set jointly by the HCP and viral-clearance constraints.

7 Process capability and robustness

Anion exchange sets the cumulative MVM clearance claim and is a major clearance step for enveloped virus (XMuLV), host-cell protein, residual DNA and leached Protein A. A Monte-Carlo simulation of 2,000 commercial-scale batches, with every parameter varied within its NOR, was used to estimate the commercial-scale distributions and process capability of the drug-substance CQAs to which this step contributes (Table 7.1). Capability is evaluated one-sided — against the lower requirement for the cumulative viral clearance and against the upper limit for the impurities.

Table 7.1: Commercial-scale process capability for the CQAs governed by the anion-exchange step.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.14
Residual DNA	VL	upper	0–0.001	0.0001 ± 0	10.81
Leached Protein A	L-M	upper	0–5	0.003 ± 0.0006	2785.30
Viral clearance — XMuLV (enveloped)	VH	lower	16.7–30	19.5 ± 0.61	1.53
Viral clearance — MVM (parvovirus)	VH	lower	8.6–20	10.4 ± 0.4	1.51

The cumulative MVM clearance meets its 8.6 log requirement with a capability of **Cpk = 1.51** (achieved cumulative 10.0 log); as the first and principal parvovirus-clearance step, anion exchange is the primary contributor to this outcome, and this is the tightest process capability of the A-Mab drug substance. The cumulative XMuLV clearance meets its 16.7 log requirement with Cpk = 1.53 (achieved cumulative 18.9 log), to which anion exchange contributes orthogonally alongside low-pH inactivation and virus filtration. The drug-substance HCP meets its 100 ng/mg limit with Cpk = 6.14, this step completing the reduction begun at Protein A and cation exchange; and the residual DNA and leached Protein A are cleared to far below their limits (Cpk = 10.8 and 2785 respectively), with anion exchange providing a major DNA reduction and modular leached-Protein-A clearance. The capability margins confirm that the NOR-bounded step contributes robustly to in-specification drug substance at commercial scale.

8 Parameter classification

Each parameter was classified on the A-Mab continuum from the demonstrated effects and the operating-region risk (Table 8.1). The rationale is as follows.

- **Load pH — WC-CPP.** The dominant factor for the flow-through-pool HCP (higher pH improves clearance) and for the XMuLV and MVM log-reduction (higher pH improves clearance), and it participates in the significant load-pH × conductivity interaction. It is reliably held within the operating region by controlled buffer preparation with in-process pH verification.
- **Equilibration/wash-1 conductivity — WC-CPP.** Significantly affects the flow-through-pool HCP (higher conductivity worsens clearance) through the load-pH × conductivity interaction. It is controlled by buffer preparation with in-process conductivity verification.
- **Load conductivity — WC-CPP.** The second factor for the XMuLV and MVM log-reduction (higher load conductivity reduces viral binding and clearance); it is controlled by conditioning the load to the target conductivity with in-process verification.
- **Protein load — WC-CPP.** Carries a credible risk to the impurity and viral load and is therefore controlled as a well-controlled CPP; in representative load material it showed no significant effect on clearance or recovery, so its acceptable range is wide. The characterization localized the load-related risk to the **charge-variant quality of the load material** (§12), which is controlled as a feed-input attribute rather than by a load set-point.
- **Operating flow rate — WC-CPP (univariate).** Controlled to preserve the minimum residence time credited for the viral-clearance claim; assessed univariately over its range with no adverse impact on clearance or yield within the range studied.

Because every parameter that could affect the flow-through-pool HCP or the viral log-reduction is controlled as a CPP within a demonstrated operating region, and none affects process performance without also bearing on a CQA, there are no key process parameters (KPP) at this step: the four multivariate parameters and the flow rate are all well-controlled CPPs.

Table 8.1: Post-characterization parameter classification.

Parameter	Class	Study
Load pH	WC-CPP	multivariate
Equil/Wash-1 conductivity	WC-CPP	multivariate
Load conductivity	WC-CPP	multivariate
Protein load	WC-CPP	multivariate
Operating flow rate	WC-CPP	univariate

9 Contribution to the control strategy

The anion-exchange step is the primary control point for the parvovirus (MVM) clearance claim and a major control point for enveloped-virus clearance and the process-related impurity load. Its contribution to the overall control strategy comprises: control of the four WC-CPPs (load pH, equilibration/wash-1 conductivity, load conductivity and protein load) to their NORs within the established operating region; control of the operating flow rate as a WC-CPP to preserve the minimum residence time credited for viral clearance; control of the **load-material acidic charge-variant (deamidation) quality** as a feed-input attribute, with an upstream neutralized-hold control to prevent the atypical deamidation identified in Deviation DEV-01; adoption of the corrected trailing-edge UV pool-collection criterion determined in Deviation DEV-02; in-process monitoring of the load pH, conductivity and UV profile and of the pool collection window; resin life-cycle control (**SOP-2008**); and release testing of the flow-through-pool HCP, residual DNA, leached Protein A and — through the validated modular claim — the viral clearance. The MVM log-reduction credited here, together with the orthogonal clearance of low-pH inactivation and virus filtration, delivers the cumulative viral-safety claim consolidated in **PCMR-001**.

10 Discussion

The characterization quantifies the parameter–response relationships for the anion-exchange polishing step, defines a robust multivariate operating region in the four well-controlled CPPs, and classifies every parameter with a data-supported rationale. The flow-through-pool HCP, XMuLV-clearance and MVM-clearance response-surface models are adequate and their behaviour is consistent with established flow-through anion-exchange chromatography and with the A-Mab prior knowledge, increasing confidence that the SDM and the viral-spiking model are representative of commercial scale. The step yield is high and robust. Two deviations were recorded and resolved without impact to the established operating region: the load-material deamidation that invalidated the first execution (DEV-01), whose root cause is confirmed by the statistical absence of the protein-load $\times$ conductivity interaction in the requalified-load DoE and which is controlled going forward as a feed-input attribute; and the slightly permissive pool-collection UV set-point (DEV-02), corrected by modelling and confirmed by verification runs because insufficient requalified load remained for a third design. The principal limitations are inherent to scale-down and small-scale spiking characterization: the absolute clearance may shift modestly at scale and over the resin life-cycle, and the credited viral clearance depends on the load conductivity and pH being maintained — which is why the operating region is defined on the qualified models, confirmed at commercial scale during Stage 2, and carried forward with in-process monitoring, resin life-cycle control and the load-material charge-variant specification. The characterization did not identify any parameter requiring the most stringent (CPP, as opposed to WC-CPP) control within the studied ranges.

11 Conclusions

The A-Mab anion-exchange polishing step is well understood and robust. The relationships between the polishing parameters and the flow-through-pool HCP, the XMuLV and MVM log-reduction and the step yield are quantified by DoE; a multivariate operating region in the four well-controlled CPPs is established; every parameter range is justified by data; and the drug-substance CQAs governed by the step meet acceptance at commercial scale, with the step being the principal contributor to the cumulative MVM (parvovirus) clearance claim (Cpk = 1.51, the tightest capability of the drug substance) and a major contributor to enveloped-virus and HCP clearance. Two deviations were recorded and resolved without impact to the operating region or the parameter classifications. The operating region, parameter classifications, proven acceptable ranges and the modular viral-clearance credit provide the basis for the Stage 2 operating ranges and the drug-substance control strategy. This report rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment.

12 Deviations from the plan

Two deviations from **PCP-008** were recorded during execution. Both were investigated, resolved and assessed for impact; neither altered the parameter classifications or the operating region established by the DoE. They are summarized in Table 12.1 and detailed below.

Table 12.1: Deviations from PCP-008.

ID	Deviation	Disposition
DEV-01	Non-representative (deamidated) load material in the first design execution	Designs invalidated and re-executed in full on requalified load; root cause confirmed by the data reported here
DEV-02	Trailing-edge UV pool-collection set-point slightly permissive in both executions	Corrected pool-stop determined by modelling and confirmed by verification runs; common-mode offset, no impact to effects or operating region

12.1 DEV-01 — Non-representative load material (deamidation); designs re-executed

During the first execution of the screening and response-surface designs, the anion-exchange load (the cation-exchange eluate) was subsequently found to carry an **elevated acidic charge-variant (deamidation) burden**, above the range of experience for representative load (the acidic-variant design-space upper bound is 40%). The elevated deamidation was traced to an out-of-trend, extended neutralized hold of the cation-exchange eluate before anion-exchange loading, which promoted asparagine deamidation and increased the proportion of the more-acidic charge variants presented to the step.

In that first execution, at the **high-protein-load / high-equilibration-wash-1-conductivity corner** of the design, the flow-through-pool HCP rose sharply above the in-process action limit and the pool purity became unacceptable, through an anomalous **protein-load × equilibration/wash-1-conductivity interaction**: the more-acidic deamidated species altered the anion-exchange binding balance so that, when the column was heavily loaded and the wash conductivity was high, impurity breakthrough into the flow-through pool increased. Because an acceptable operating region could not be defined on a non-representative load, the affected screening and response-surface designs were **invalidated for operating-region definition**. A representative, requalified load — with the acidic charge variants within the 18–40% range of experience and the upstream neutralized-hold time restored to its normal range — was prepared, and the screening and response-surface designs were **re-executed in full**. The re-executed designs are the ones analyzed and reported in this document.

Root-cause confirmation from the reported data. In the requalified-load DoE reported here, the protein-load × equilibration/wash-1-conductivity interaction is **statistically absent** (effect 0.22 ng/mg, $p = 0.86$), and the protein load has no significant main effect on the flow-through-pool HCP, whereas the load-pH × conductivity interaction that physically governs HCP clearance is strong and significant (effect -6.6 ng/mg, $p < 0.001$). The anomalous interaction seen in the first execution was therefore an **artifact of the degraded (deamidated) load**, not an intrinsic property of the step. This finding identifies the **charge-variant quality of the load material** as the controlled input that must be assured — by a feed-input acidic-variant specification and by upstream neutralized-hold control — rather than by an anion-exchange set-point, and it is carried into the control strategy (§11). No product was released against the invalidated execution; the characterization conclusions rest solely on the requalified-load designs.

12.2 DEV-02 — Pool-collection UV set-point corrected by modelling and verification runs

Both executions collected the flow-through/wash pool to a fixed trailing-edge UV (280 nm) pool-stop that was subsequently found to be set **slightly high (permissive)** relative to the optimum — the pool was collected marginally further down the descending wash than ideal, so a small, approximately constant amount of late-breakthrough HCP was carried into every pool.

Impact assessment — the offset is common-mode. Because the identical UV set-point was applied to every run, its effect is a **common-mode offset**: it shifts the absolute flow-through-pool HCP level (and, negligibly, the yield) by an approximately constant amount but does not change the factor effects, the two-factor interactions or the geometry of the operating region — adding a constant to every response changes only the model intercept, not the coefficients. The parameter classifications and the design-space geometry reported here are therefore **unaffected** by the pool-stop setting; only the absolute in-process pool-HCP set-point and the manufacturing pool-collection method setting required correction. Even under the original (slightly permissive) setting, the flow-through-pool HCP at the set-point (14 ng/mg) sits far below the   100 ng/mg drug-substance limit, so the step met its purpose throughout; the correction improves the margin and fixes the validated manufacturing pool-stop.

Determination by modelling. The requalified load was **exhausted** by the re-executed screening and response-surface designs, and insufficient material remained for a third full design. The corrected pool-stop was therefore determined by **modelling rather than by re-running the DoE**. The chromatographic UV traces and the fraction-resolved pool analytics recorded during the re-executed response-surface runs were used to reconstruct the cumulative flow-through-pool HCP and product mass as functions of the pool-stop UV threshold — a trailing-edge model in which lowering the threshold excludes the last, most-impure wash fractions, reducing the pool HCP at a small yield cost. The model, fitted on the set-point and centre-point runs, predicts the pool-stop UV threshold that **minimizes the flow-through-pool HCP subject to maintaining step yield   the target**; the predicted optimum is a **tighter (lower) UV threshold** than the one used in the DoE, reducing the modelled set-point pool HCP below the level collected under the original setting at a negligible predicted yield cost (well within the flow-through step's recovery margin, nominal yield 97.3%).

Verification runs — settings and number. The modelled optimum was confirmed with a small, material-frugal set of verification runs, designed as follows.

- **Objective.** Confirm that at the corrected pool-stop (i) the set-point flow-through-pool HCP and step yield match the model prediction within the pre-defined tolerance, and (ii) the corrected threshold holds at the worst-case corner of the operating region for pool HCP.
- **Settings.** The corrected pool-stop UV threshold, with the four chromatographic factors at their set-points for the confirmation runs, and at the **high-load / high-equilibration-wash-1-conductivity corner** (the response-surface-identified worst case for flow-through-pool HCP) for the robustness runs; all other conditions per **SOP-2011**.
- **Number — three confirmation runs plus two robustness runs (five in total).** The number is justified statistically rather than by convention:
  - *Three confirmation runs* at the corrected pool-stop and nominal set-points place a two-sided 95% confidence interval on the mean set-point flow-through-pool HCP whose half-width, using the response-surface centre-point pure-error standard deviation (s = 2.5 ng/mg, %CV = 18%), is 6.2 ng/mg — only 7% of the large margin between the set-point pool HCP (14 ng/mg) and the 100 ng/mg drug-substance limit — which is sufficient to compare the observed mean against the model's 95% prediction interval and to demonstrate control with margin.
  - *Two robustness runs* at the worst-case corner confirm that the corrected threshold still yields acceptable flow-through-pool HCP where the response surface predicts the pool HCP is highest, testing the correction at the edge rather than only at the centre of the region.
  - A *full third response-surface design* (28 runs) was **not** justified: the pool-stop correction is a common-mode offset on an already-established design-space model, not a new factor, so re-running the design would consume the exhausted requalified load without adding operating-region information. Five verification runs are the minimum sufficient to verify the modelled set-point and its worst-case robustness.
- **Outcome.** The verification runs met the pre-defined acceptance criteria: the observed set-point flow-through-pool HCP fell within the model's prediction interval and well below the drug-substance limit, the corrected pool-stop reduced the flow-through-pool HCP relative to the original setting with no material yield loss, and the worst-case runs confirmed acceptable flow-through-pool HCP at the edge of the operating region. The corrected trailing-edge UV pool-collection criterion is adopted as the manufacturing pool-stop and carried into the control strategy (§11); it does not alter the parameter classifications or the operating region established by the DoE.

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Appendix A — Screening design matrix and responses

The coded two-level full-factorial design (factors A–D per Table 4.2; coded levels −1/0/+1) and the measured responses for each run.

Table 12.2: Appendix A1 — screening design (coded factor settings).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

Table 12.3: Appendix A2 — screening run responses.

Run	Pool HCP (ng/mg)	XMuLV LRF (log)	MVM LRF (log)	Step yield
1	12.54	6.159	4.587	0.9898
2	15.65	6.149	4.431	0.9749
3	12.11	5.473	3.756	0.9506
4	14.55	5.563	3.962	0.9642
5	35.68	6.335	4.738	0.9655
6	35.05	6.301	4.826	0.995
7	29.19	5.924	4.009	0.9742

Run	Pool HCP (ng/mg)	XMuLV LRF (log)	MVM LRF (log)	Step yield
8	29.81	5.952	3.943	0.969
9	9.618	7.005	5.915	0.973
10	7.371	7.396	5.611	0.9532
11	8.685	6.613	5.25	0.9724
12	8.036	6.709	4.895	0.9727
13	12.68	6.593	5.912	0.995
14	14.96	8.341	5.971	0.9808
15	12.86	6.685	5.146	0.9656
16	15.03	6.522	5.315	0.9652
17	16.16	6.42	4.539	0.9655
18	17.53	6.796	5.075	0.9748
19	11.59	5.946	4.826	0.9743

Appendix B — Response-surface design matrix and responses

The coded face-centred central-composite design (factors A–D; coded levels $-1/0/+1$) and the measured responses for each run.

Table 12.4: Appendix B1 — response-surface design (coded factor settings).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

Table 12.5: Appendix B2 — response-surface run responses.

Run	Pool HCP (ng/mg)	XMuLV LRF (log)	MVM LRF (log)	Step yield
1	13.54	6.722	4.546	0.9508
2	17.3	6.027	5.212	0.9891
3	14.06	5.865	4.019	0.995
4	15.04	5.297	3.762	0.9511
5	28.52	6.527	4.773	0.9875
6	25.46	6.714	4.713	0.9634
7	31.26	5.356	3.908	0.995
8	32.27	5.064	3.359	0.966
9	5.663	7.498	6.142	0.9744
10	8.366	7.827	5.613	0.9886
11	7.555	6.493	4.539	0.9722
12	7.431	7.13	4.676	0.9881
13	17.61	7.179	6.2	0.9738
14	18.04	8.123	6.33	0.9683
15	13.86	6.582	4.532	0.9596
16	13.85	6.37	5.159	0.9735
17	23	5.835	4.099	0.9707
18	12	7.374	4.912	0.9647
19	8.105	6.418	4.932	0.9459
20	27.52	5.996	5.02	0.972
21	15.63	7.441	5.139	0.9524
22	15.52	5.934	4.56	0.9675
23	12.98	6.518	4.811	0.9683
24	14.25	6.704	5.115	0.9521
25	14.97	6.591	4.421	0.9576
26	10.74	6.025	5.197	0.9748
27	16.34	7.049	5.045	0.9774
28	12.62	6.856	4.919	0.9527

Appendix C — Full effect and coefficient tables

12.3 Screening effects — all responses

Pool HCP (ng/mg)

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	12.1	6.04	0.613	9.85	9.48e-06	***
A	-11.9	-5.96	0.613	-9.72	1.05e-05	***
AB	-6.63	-3.32	0.613	-5.41	0.000642	***
C	-1.66	-0.829	0.613	-1.35	0.213	
AC	1.65	0.827	0.613	1.35	0.214	
BC	-1.21	-0.605	0.613	-0.986	0.353	
D	0.885	0.443	0.613	0.722	0.491	
AD	-0.498	-0.249	0.613	-0.406	0.696	
CD	0.261	0.13	0.613	0.212	0.837	
BD	0.224	0.112	0.613	0.182	0.86	

XMuLV LRF (log)

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1	0.501	0.0982	5.1	0.000934	***
C	-0.605	-0.302	0.0982	-3.08	0.0151	*
D	0.268	0.134	0.0982	1.37	0.209	
CD	-0.255	-0.128	0.0982	-1.3	0.23	
AD	0.25	0.125	0.0982	1.27	0.24	
B	0.198	0.0991	0.0982	1.01	0.342	
BD	0.126	0.0632	0.0982	0.643	0.538	
AC	-0.0968	-0.0484	0.0982	-0.493	0.635	
AB	-0.094	-0.047	0.0982	-0.478	0.645	
BC	-0.0169	-0.00844	0.0982	-0.0859	0.934	

MVM LRF (log)

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.22	0.61	0.0446	13.7	7.83e-07	***
C	-0.715	-0.357	0.0446	-8.01	4.32e-05	***
B	0.182	0.091	0.0446	2.04	0.0757	

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
BD	0.107	0.0536	0.0446	1.2	0.263	
AD	-0.0627	-0.0314	0.0446	-0.704	0.502	
D	-0.0451	-0.0226	0.0446	-0.506	0.626	
BC	-0.0442	-0.0221	0.0446	-0.496	0.634	
CD	0.0334	0.0167	0.0446	0.375	0.718	
AC	0.0135	0.00677	0.0446	0.152	0.883	
AB	-0.0132	-0.00659	0.0446	-0.148	0.886	

Step yield

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
C	-0.0116	-0.00582	0.00315	-1.85	0.102	
B	0.00743	0.00371	0.00315	1.18	0.272	
AD	-0.00715	-0.00357	0.00315	-1.13	0.29	
AC	0.00513	0.00257	0.00315	0.815	0.439	
BC	-0.00392	-0.00196	0.00315	-0.622	0.551	
BD	0.00383	0.00192	0.00315	0.608	0.56	
CD	0.00346	0.00173	0.00315	0.549	0.598	
AB	0.00141	0.000707	0.00315	0.224	0.828	
D	-0.00136	-0.000679	0.00315	-0.215	0.835	
A	-0.000663	-0.000332	0.00315	-0.105	0.919	

12.4 Response-surface coefficients — XMuLV log-reduction and step yield

XMuLV LRF (log)

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	6.57	0.109	60.6	2.51e-17	***
A	0.621	0.074	8.39	1.33e-06	***
B	-0.076	0.074	-1.03	0.323	
C	-0.554	0.074	-7.48	4.6e-06	***
D	0.0287	0.074	0.388	0.704	
AB	-0.0278	0.0785	-0.355	0.729	
AC	0.0223	0.0785	0.285	0.78	
AD	0.192	0.0785	2.44	0.0295	*
BC	-0.118	0.0785	-1.5	0.158	
BD	0.0577	0.0785	0.735	0.475	
CD	-0.0751	0.0785	-0.957	0.356	
A^2	0.0696	0.195	0.356	0.727	
B^2	-0.328	0.195	-1.68	0.118	
C^2	0.152	0.195	0.779	0.45	

Term	Coef.	Std. err.	t	p-value	Sig.
D^2	0.0761	0.195	0.39	0.703	

Step yield

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.961	0.00501	192	7.92e-24	***
A	-0.000294	0.00342	-0.086	0.933	
B	0.00022	0.00342	0.0643	0.95	
C	0.00109	0.00342	0.318	0.756	
D	-0.00203	0.00342	-0.594	0.563	
AB	-0.00464	0.00362	-1.28	0.223	
AC	-0.00175	0.00362	-0.482	0.637	
AD	0.00608	0.00362	1.68	0.117	
BC	-0.000152	0.00362	-0.042	0.967	
BD	-0.00433	0.00362	-1.2	0.253	
CD	-0.00413	0.00362	-1.14	0.275	
A^2	0.00932	0.00902	1.03	0.32	
B^2	0.000524	0.00902	0.0581	0.955	
C^2	0.00152	0.00902	0.168	0.869	
D^2	0.00177	0.00902	0.196	0.847	

Appendix D — Analytical methods summary

Table 12.6: Analytical methods, techniques and validation references (performance characteristics per the cited method validation reports).

Method	Technique	Analytes / attributes	Validation ref.	Reportable
Host-cell protein	ELISA	HCP	AMV-3012	ng/mg
Residual DNA	qPCR	Host-cell DNA	AMV-3014	ng/dose
Leached Protein A	ELISA	Leached Protein A	AMV-3016	ppm
XMuLV clearance	Infectivity titre (TCID50)	Enveloped model virus	AMV-3017	log LRV
MVM clearance	Infectivity titre (TCID50/qPCR)	Parvovirus model virus	AMV-3018	log LRV
Charge variants	icIEF	Acidic / main / basic variants	AMV-3013	% acidic